TERMINI · AI BUILDER CUP 2026 · FUTURE OF WORK & ENTERPRISE PRODUCTIVITY

Termini

Every organisation should be able to run its own intelligence. **Termini is the layer that makes that possible** — bare hardware in, a governed AI fleet out, and the data never leaves the building.

“Your hardware. Your data. Our operations.”

Day-0 · registered in 150 ms

Day-2 · self-healing, audited

Fail-closed governance

01 THE CASE

One clinic, one rule

A dental clinic in Korea wanted AI for its phones, its insurance claims and its own reporting. Sending patient data to a cloud was not an option — so the owner, a biology graduate rather than a computer scientist, built the infrastructure himself.

> **All client data must not leave the building.**
>
> The rule the whole build was organised around

6 months

of evenings, on top of dentistry — **the cost of there being no product**

0

client records sent off-site. Phones, claims and reporting all run in the back room

Source: arca.live/b/aihomelab/185810092 — “A neighbourhood dental clinic's AX transformation”, 13 Oct 2026.

02 THE CONSTRAINT

What can't leave the building

The clinic's rule is not a dental rule. It is the rule of every business whose most valuable data is also its most restricted.

Healthcare & dental
Patient records, call recordings, insurance claims
→ Front desk, adjudication, clinical notes

Banking & insurance
KYC files, transaction histories, loss records
→ Fraud triage, credit memos, claims

Legal
Privileged matter files, client advice
→ Contract review, discovery, drafting

Manufacturing
Design files, process parameters, yields
→ Quality, maintenance, planning

Life sciences
Trial data, patient histories
→ Protocol analysis, safety review

Public sector & defence
Classified material, citizen records
→ Analysis, briefings, casework

Software
Source code, incident data, architecture
→ Code agents, on-call triage

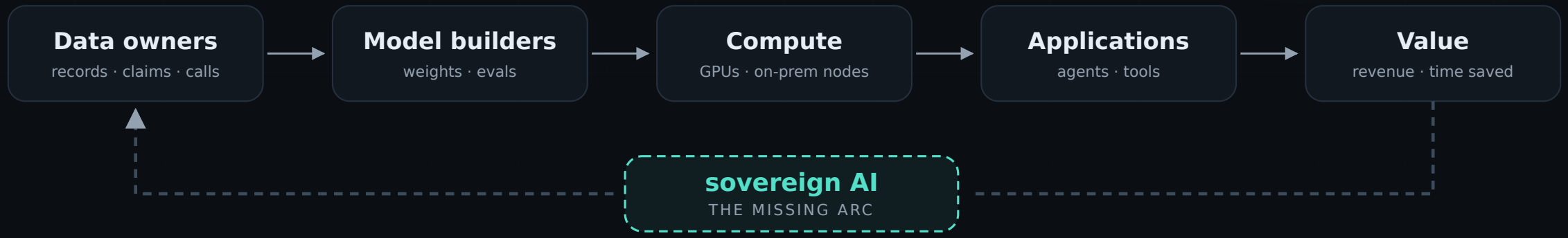
Retail & commerce
Customer PII, supplier terms, margins
→ Demand, service, pricing

Eight industries, one constraint: **the data that would make AI useful is the data that can't be handed over.**

And the result: **95%** of GenAI pilots with no measurable P&L impact (MIT NANDA 2025, contested) while **88%** of firms use AI somewhere (McKinsey 2025) — permission, operations and context are missing at once.

The loop only closes where the data lives

A system improves by learning from its own outcomes — record, decide, act, observe, record again.
When the record lives in someone else's cloud, the loop belongs to someone else.



A memory that stays

Outcomes, corrections and exceptions remain with the entity, so every decision is measured against the last one.

A jury, not an oracle

Several models read the same evidence. Agreement confirms; disagreement is where a person looks.

A corpus no one else has

The private record — claims, calls, outcomes — that no foundation model has seen and no vendor could be given.

Intelligence should be plural. **Governance is not the tax on the loop — it is what closes it.**

04 UNAUTHORISED USE

Data used without permission

The loud concern is AI replacing existing labour. The governing one arrives first: the work leaves the building, and comes back as someone else's product.

APR 2023 · SAMSUNG

Engineers debugged with ChatGPT

Confidential **source code**, equipment yield data and a full meeting's notes went in as prompts. Samsung restricted generative AI across its network weeks later.

"Data is transmitted and stored to an external server, making it impossible for the company to retrieve it." — Samsung internal memo

JAN 2025 · DEEPSEEK

A database of everyone's prompts

Wiz Research found an unauthenticated ClickHouse instance holding **1M+ log lines**, chat history, API keys and plaintext passwords.

Not a breach of the model — a copy of every conversation, reachable from the open internet.

SEP 2026 · NAVIER-STOKES

The credit left with the data

OpenAI announced a Navier–Stokes solution that its own team described as using **"very little human input"**. Twelve hours earlier, Buckmaster said his and Alpöge's unpublished Euler advances had leaked to the lab.

A Millennium Prize result — and a priority dispute about whose work the model had read.

HOW TERMINI CLOSES IT Not a better policy — a topology where **the copy never leaves**: C2 stays on the node, and what does leave is named, classified and audited.

Termini turns hardware you own into AI you can govern

Every use case so far needs the same three things, and no vendor sells them together. The clinic needed all three, so one man built all three by hand. In Termini they are one product.

01 · BOOTSTRAP

Hardware becomes a fleet

A bare node boots the pre-baked image, registers in **150 ms**, is assigned a role and joins the cluster. No installs, no specialist on site.

Removes: the wiring weekend

02 · OPERATE

Self-healing, not alerting

When memory or disk crosses a limit, Termini restarts the runaway workload, records why it chose that action and what happened, and falls back to a fixed rule whenever the model is unsure.

Removes: the on-call rotation

03 · PROVE

A boundary you can audit

C0 / C1 / C2 classification, an on-node redactor and a **fail-closed** controller gate. What leaves is named, classified and logged.

Removes: the compliance gap

The clinic is the requirement. **Termini is the product.**

Right-size the intelligence

Frontier models are not the answer to every task. They are the top of a ladder — and the ecosystem is the routing between the levels.

LEVEL	RUNS ON	GOOD AT	DATA MOVEMENT	IN THE DEMO
L0 DETERMINISTIC	Controller rules	Known remedies, threshold-to-action mappings	None	**Yes**
L1 SMALL LOCAL	The node — 4B class, `llama.cpp`	Classify, extract, summarise, triage	None	**Yes**
L2 LARGE LOCAL	A GPU node over your own corpus	Reasoning over your own records	None	**Config only**
L3 FRONTIER	Gemini, via Vertex AI	The hardest reasoning, the smallest payload	Minimised C0, if permitted	**Yes**

L1 triages → **L3 reasons — only where policy allows** → **L0 executes** → **The audit records which level answered**

HOW TERMINI CLOSES IT The ladder is a routing table; the boundary decides what may climb it. **Local is not worse** — it is the right level for most work.

Six months, compressed into an afternoon

Bare node → Pre-baked image → Registered 150 ms → Role assigned → Agents running

	THE DENTIST'S PATH	TERMINI'S PATH
Hardware	Wiring, hypervisor, cluster join, PBX — by hand, at night	Boot the image; the controller registers it and joins the cluster
Model	Pick, quantise, serve, benchmark, re-pick	A config knob — Gemma resident, Gemini where the customer permits it
Operations	Watch it yourself, forever	A governed operator plus a human dashboard
Governance	A rule enforced by discipline	A gate that rejects C2 fail-closed, before storage

What took him six months of evenings is **the product's first hour**.

Lightweight teams, heavy infrastructure

A small team should not have to hire infrastructure engineers to move quickly. The infrastructure should arrive already operating.

Telemetry → **Threshold rule fires** → **Model decides — allowlist only** → **Audited LXD execution** → **Audit row**

1 It watches

Thresholds on memory, disk and inference health. The loop runs **whether or not anyone is awake**, and a rule — never a model — is what notices.

2 It decides, inside a cage

The model chooses among **allowlisted actions**; grammar-constrained decoding makes anything else unencodable, and the deterministic fallback is always available.

3 It leaves a receipt

Every autonomous change links **trigger → decision → action → result** in one audit record, so a small team can prove what happened without having watched it.

The small team, **with serious infrastructure behind it.**

09 HOW IT LANDS

Delegate the operations, not the data

That split is the whole proposition. Packaging, partners and pricing all follow from it — and the boundary is what makes the delegation a contract rather than a hope.

A What is delegated

Bootstrap, remediation, upgrades, the on-call rotation. **The operations.** This is the work a twelve-person firm cannot staff and a hardware owner does not want to build.

B What is never delegated

C2 payloads, transcripts, records, credentials. **The data.** It stays on the node, and the controller sees only what the classification allows.

C Who buys which half

The **SME** buys the outcome. The **hardware owner** licenses the layer so its metal is sellable as AI. Both get the same control plane.

One control plane, two buyers, one boundary — **and the boundary is what makes delegation signable.**

What runs today, measured

150 ms

Node registration from the pre-baked image — **zero installs at boot**

0

C2 payloads the controller has ever stored — fail-closed by design

422

C2 field **rejected before storage** — fail-closed, three ways checked

1

Remediation **decided by the model**, executed, audited end-to-end

1.8 s

Diagnosis warm on the managed path — **402 s** on local CPU weights, stated not hidden

live

On GCE + Firebase + Vertex AI — **a judge can open it and watch**

Three claims, three measurements. **Nothing on this slide is a projection.**

The architecture is the governance claim

Node on premises → **On-node redactor** → **Outbound-only mTLS** → **Controller gate** → **Operator loop** → **Audit chain**

Substrate and channel — adopted, not invented

Ubuntu and an **LXD cluster** across nodes; one **outbound-only mTLS WebSocket** per node carrying heartbeat, telemetry, commands and acks. No inbound path from the internet, and no route from an agent VM to the controller.

Boundary and loop — the two things we build

A **fail-closed** telemetry gate that rejects a non-conforming batch rather than dropping fields, and a **governed operator loop**: detect → decide inside an allowlist → execute against LXD → write the audit row.

EU AI Act · Art. 10 / 12 / 14 / 15 · **GDPR** · Art. 25 / 32 / 44–49 · **ISO/IEC 42001** · 27001 · **SOC 2** Type II · **HIPAA** §164.312

The node keeps the data. The controller keeps the proof — which rule fired, which model answered, which action ran, what left the boundary — **evidence an auditor can read without touching a record.**

We don't certify you. We generate the evidence a certification asks for, continuously. High-risk obligations under the AI Act apply from 2 Dec 2027.

What leaves the building, exactly

CLASS	LEAVES?	CONTENTS
C0	Yes — allowlisted	Hardware and OS health: CPU, RAM, disk, GPU, inference liveness, instance inventory, rule metrics
C1	Yes — transformed	Salted hashes, bounded display labels, timestamps. Never raw identifiers
C2	**Never**	Workload payloads, call transcripts, patient or customer records, prompts, credentials, logs

The one external call — disclosed

The operator's **diagnosis** runs on Gemini via Vertex AI from the controller's loopback listener. Prompt: **C0 telemetry only**. The audit row records `egress=external`. One variable, `backend=local`, puts the same decision back on the node's own weights.

Why we publish the exception

Because a boundary you cannot see is not a boundary. The claim is not “nothing leaves” — it is **“nothing C2 leaves, and everything that does is named, classified and audited.”**

Gemini decides. Gemma stays home.

1 Gemini 2.5 Flash · Vertex AI

The **diagnosis** — naming the remediation from C0 telemetry. Measured 1.8 s warm. The only call that leaves the premises, and it says so in the audit trail.

2 Gemma 3 4B · on Node C

Automatically detects the GPU node, bootstraps the inference toolchain and leaves it **ready for agents to use** — `llama.cpp`, open-weight models, nothing forwarded.

3 Cloud Run · GPU reference

The same Gemma on **1× L4** with vLLM, deployed as the managed reference model — and the shape this takes when GPU quota stops being the blocker.

GCE · controller · **GCE · node C** · **Firebase Hosting · dashboard** · **Cloud Storage · weights** · **Pre-baked node image**

Local data uses local models. Where the customer permits it, **Termini integrates with Gemini.**

14 WHAT'S NEXT

From one clinic to a fleet of them

Real bare-metal onboarding

MAAS + iPXE netboot, so a box arrives from a partner and joins without a technician on site.

A GPU node in the demo

L4 quota is the current blocker; the path is already built and served on Cloud Run.

Learned baselines

Edge anomaly detection replacing static thresholds — the operator learns each site's normal.

Federated eval exchange

Evals and weights travel between entities; records don't. The plural-ecosystem mechanism, productised.

Compliance mapping

GDPR, EU AI Act, Korea PIPA, India DPDP posture mapped onto the C0/C1/C2 boundary.

Hardware partner channel

The box is commodity. The operator is the product. Partners sell metal; we run it.

The long tail does not need a data centre. **It needs one box, one afternoon, and someone else on call.**

TERMINI · AI-INFRA

The doctor built it alone. The next one won't have to.

Lightweight teams. Heavy infrastructure. Bare hardware in, a governed AI fleet out.

“Your hardware. Your data. Our operations.”

Live demo | Architecture | Questions